

Darsh Vaghasia

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CERTIFICATION

[Azure Data Scientist Associate](#) | [AWS Certified: Cloud Practitioner](#) | [Data Engineer Associate](#) | [Machine Learning Specialty](#)

SKILLS

Apache Hadoop, Apache Spark, Apache Kafka, Python, PySpark, Redshift, Snowflake, Airflow, GCP, AWS Glue, PostgreSQL, MongoDB, Cassandra, PowerBI, Jenkins, SQL, BigQuery, Azure, AWS, Microsoft Azure, NoSQL, Gitlab, STAR Schema, ETL, ELT, Talend

WORK EXPERIENCE

[Encrypt My Work](#) | **Software Engineer**, Cranford, NJ (8 Months)

Jun 2024 - Present

- Developed a scalable **Python Django** web application on **GCP** using **App Engine**, with a front-end built using **HTML** and **JavaScript**.
- Designed and optimized a **PostgreSQL** database via **Cloud SQL** for efficient data management. Conducted **A/B testing** to validate functionalities and refine new features. Automated daily user login log extraction using Shell scripting, reducing manual efforts.
- Integrated **Stripe API** for managing subscription-based SaaS products and ensured zero-downtime migration of active subscriptions.
- Automated invoicing and payment retry, improving efficiency, and resolved user queries and technical issues promptly.
- Implemented an **AI-powered RAG** support system with the **Llama** model to deliver precise answers to product-related queries.
- Worked with **Twilio** to implement secure email and OTP-based authentication, enhancing user account security and trust.
- Leveraged **Looker** and **SQL** to analyze customer data, delivering actionable insights and identifying KPIs to boost sales.
- Delivered a robust application that enhanced user experience, streamlined processes, and enabled data-driven decision-making.

Reliance Jio Platforms Limited | **Data Scientist**, Mumbai, India (6 Months)

Jan 2022 – Jun 2022

- Analyzed IoT time-series data from the Boiler Feed Water system to predict failures and identify preventive maintenance prospects.
- Conducted **EDA**, preprocessed data (scaling, feature engineering), and developed a **Multi-Scale Convolutional Recurrent Encoder-Decoder** model for anomaly detection, leveraging **ConvLSTM** to capture inter-sensor correlations and temporal patterns.
- Achieved **95% precision**, identifying root causes of anomalies and enabling preventive measures that improved system uptime.

Manentia Advisory Pvt LTD | **Machine Learning Engineer**, Vadodara, India (5 Months)

Aug 2021 – Jan 2022

- Preprocessed and augmented retinal fundus images (88,702 images) using noise removal, normalization, and morphological contrast enhancement techniques to improve data quality and model performance. Streamlined feature extraction using the green channel and edge detection methods, enabling early diagnosis of subtle abnormalities like microaneurysms in retinal images.
- Developed and trained a CNN model with enhanced activation functions for Diabetic Retinopathy detection, achieved 99.75% precision.

Alma Mater Biz Solutions PLC | **Data Analytics & Software Developer Intern**, Ahmedabad, India (3 Months)

Apr 2020 – Jul 2020

- Orchestrated an E-commerce website on Amazon EC2 using WordPress, ensuring efficient data management with AWS RDS.

ACADEMIC PROJECTS

[AWS CI/CD Automation using Gitlab](#) (Gitlab, Docker, AWS S3, Elastic Beanstalk, CI/CD)

Sep 2024 – Oct 2024

- Streamlined **CI/CD** with GitLab for Dockerized app deployment to **AWS S3** and **Elastic Beanstalk**, utilizing DevOps best practices.
- Developed a GitLab CI/CD pipeline** to build, test, and deploy securely and efficiently, ensuring early error detection. Achieved seamless deployment and enhanced development processes, resulting in improved efficiency and security for the application.

[Crick Forge: Automated Data Architecture for Match Insights](#) (Snowflake, SQL, Stream, DAG)

Apr 2024 – May 2024

- Designed **Snowflake**-based data pipeline for **Delta Data load** of nested **JSON** cricket data, from schema setup to analytics-ready storage.
- Implemented advanced data transformation techniques for detailed **fact** and **dimension** tables, enhancing match analysis.
- Automated end-to-end workflow with **DAG's** and Snowflake's **Stream** and **Task** objects achieving **60%** speed improvement in seamless data ingestion, processing, and visualization generation for enhanced cricket match insights.

[Real-time AWS-Snowflake Integration](#) (AWS, Snowflake)

Apr 2024 – May 2024

- Spearheaded real-time data pipeline on **AWS**, orchestrating hourly data ingestion from Weather API to **DynamoDB**, then **Snowflake**.
- Engineered fault-tolerant data flow, utilizing **DyanamoDB** streams and **AWS Lambda** for real-time processing, and reliable data loading into **Snowflake** from **AWS S3**. Integrated AWS Services and Snowflake, enabling scalable data transfer and analysis, enhancing decision-making with actionable insights.

[Data Pipeline Optimization and Automation Framework](#) (Azure, PySpark, Cloudera, SQL)

Jan 2024 – Feb 2024

- Designed and implemented a scalable ETL pipeline using **PySpark** and **SQL** for a Cloudera-based system, optimizing data transformation.
- Leveraged Azure Data Factory for orchestration and Databricks for data processing, enabling seamless data integration.

EDUCATION

M.S, Computer Science Specialization in Data Science | **The University of Texas at Dallas, TX**

Aug 2022 - Aug 2024

Courses: Machine Learning, Big Data MGMT and Analytics, Database Design, Statistical Methods for Data Science, Design Analysis and Computational Algorithm, Data Science for Smart Cities, Virtual Reality, Secure Cloud Edge & IOT, Algorithm Analysis and Data Structures

B. Tech, Computer Science & Engineering | **Charotar University of Science and Technology**

Jul 2018 - May 2022

Courses: Artificial Intelligence, Digital Image Processing, Data Science & Analytics, Cloud Computing, Theory of Computation, Database Management